ABHISHEK SHARMA

AI / ML ENGINEER

+91 78766 30851 • abhis.ds01@gmail.com • LinkedIn • Hamirpur, Himachal Pradesh, India

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building Retrieval-Augmented Generation (RAG) chatbots, agentic LLM workflows (LangChain, LangGraph, OpenClaw), computer-vision pipelines, and full-stack ML-integrated web applications in Python, Django, and TensorFlow. Skilled in parameter-efficient fine-tuning of LLMs (LoRA/QLoRA), fine-tuning object-detection models (YOLO), tracing and evaluating LLM apps with LangSmith, automating GitHub-based project ingestion, and shipping production features on Streamlit and Firebase. Experienced translating client requirements into working AI solutions end-to-end — from JD-matched proposal generation to secure, database-backed deployment.

TECHNICAL SKILLS

Programming: Python, SQL
AI / ML: Machine Learning, Deep Learning, NLP, Computer Vision, Object Detection (YOLO), Image Segmentation (SAM), Depth Estimation (Depth Anything), Recommendation Systems
Generative AI: LLMs, Retrieval-Augmented Generation (RAG), AI Chatbots, Prompt Engineering, Replicate API, Claude Code
LLM Frameworks & Agents: LangChain, LangGraph (stateful multi-step agent workflows), LangSmith (tracing, evaluation, debugging), OpenClaw agent integration, Tool/Function Calling
LLM Fine-Tuning: LoRA, QLoRA, Parameter-Efficient Fine-Tuning (PEFT), 4-bit Quantization, Hugging Face Transformers
Frameworks & Libraries: TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, BeautifulSoup
Web & Backend: Django, REST API, WebSocket, Streamlit, Firebase
Cloud & DevOps: AWS (EC2, Lambda, S3), Docker, Celery, Redis
Testing & Automation: Playwright, GitHub Repo Crawling/Automation, Bitrix24 Webhooks
Databases & Security: MySQL, PostgreSQL, SQLite, MySQL Workbench, Data Hashing & Security
Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab

EXPERIENCE

AI/ML Engineer | *Devexhub Private Ltd., Mohali, Punjab* *Feb 2026 – Present*

- Design and ship production AI features spanning generative AI, agentic LLM workflows, computer vision, and full-stack automation for internal and client-facing use cases.
- Own end-to-end delivery across four major internal platforms — from data pipelines and model integration to secure AWS deployment (EC2, Lambda, S3) using Docker, Celery/Redis, and Playwright (see Projects).
- Partner with sales teams to convert client job descriptions into technical proposals backed by matched case studies from company project history.

Python & AI/ML Intern | *Sunfocus Pvt. Ltd.* *Jun 2025 – Jan 2026*

- Automated repetitive backend tasks using Python and SQL queries, cutting manual processing effort.
- Built interactive Streamlit web applications and dashboards for internal data workflows.
- Used NumPy and Pandas to clean, analyze, and manipulate project datasets.

PROJECTS

AI Knowledge Chatbot & Proposal Automation *Python, RAG, LangChain, LangGraph, LangSmith, LLMs, GitHub API*

- Built a Retrieval-Augmented Generation (RAG) chatbot with LangChain that ingests the company's entire codebase and project history to answer technical and business queries in real time.
- Designed a LangGraph agent workflow orchestrating retrieval, tool calls, and response generation as stateful steps, and integrated OpenClaw agents to automate multi-step tasks around the chatbot.
- Instrumented the pipeline with LangSmith to trace runs, debug retrieval quality, and evaluate responses across prompt and model changes.
- Engineered an automated GitHub crawling pipeline that generates a structured project profile directly from a given repository URL, eliminating manual documentation work.
- Developed an AI-assisted proposal workflow that matches an incoming client job description against the project database and auto-populates relevant case studies into the final proposal.
- Powered the chatbot and proposal workflow with free/open-source LLMs instead of paid APIs, cutting recurring inference costs while maintaining response quality; applied LoRA/QLoRA fine-tuning to adapt open-source models efficiently on limited GPU memory.

HR Management System *Django, PostgreSQL, Bitrix24*

- Built a full HR platform covering onboarding, exit tracking, salary processing, and employee training, used company-wide.
- Created a themeable document-generation engine producing offer letters, appointment letters, and salary slips.
- Integrated the platform with Bitrix24 via a webhook-based pipeline to sync employee data in real time, and secured salary/PII records with database-level hashing.

Basketball Performance Analytics *YOLO, Deep Learning, Computer Vision*

- Fine-tuned a YOLO object-detection model on basketball game footage to track players and the ball in real time.
- Applied deep learning post-processing to derive player- and team-level performance statistics from raw video.

AI Product Mockup Generator *YOLO, SAM, Depth Anything, Replicate API*

- Built a pipeline that detects an object (pen, bottle, bag, etc.), segments its surface, and estimates depth to place a logo/wrap at correct scale and perspective.
- Combined YOLO, SAM, and Depth Anything with the Replicate API to generate photorealistic product mockups automatically.
- Used Replicate's pay-per-use GPU inference instead of provisioning dedicated GPU infrastructure, lowering compute costs for on-demand generation.

EDUCATION

Bachelor of Computer Application (BCA) — 73% | *Swami Vivekanand Govt. Utkrisht College, Ghumarwin* *2022 – 2025*
Senior Secondary (12th, PCM) — 78% | *B.B.N.S.S. School, Chakmoh* *2021 – 2022*
Matriculation — 88% | *Government School, Chakmoh* *2019 – 2020*
Interests: Badminton, Cooking